

HR ANALYTICS

Wireframe Documentation

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1. Introduction

1.1 Purpose of the Document

The Wireframe Design Document outlines the proposed user interface and layout for the HR Analytics project. Wireframes serve as a blueprint for the application, illustrating how users will interact with the system and how information will be organized. This document aims to provide clarity on design decisions and help stakeholders visualize the application before development begins.

1.2 Overview of the HR Analytics Project

The HR Analytics project aims to analyze employee data to predict attrition. The application will allow HR professionals to:

- Input and manage employee data.
- Perform data analysis to uncover trends related to employee satisfaction and turnover.
- Generate comprehensive reports to facilitate data-driven decision-making.

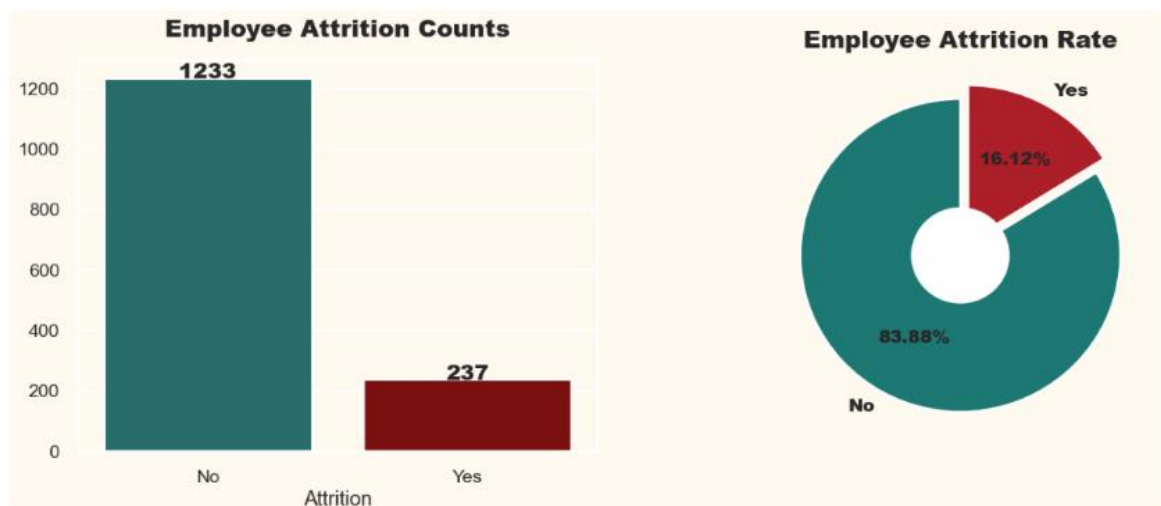
2. Wireframes

2.1 Home Page Wireframe

Description: The home page serves as the primary dashboard for the HR Analytics application, providing users with a quick overview of key metrics and navigation options.

Key Elements:

- **Header:**
 - Application logo and title ("HR Analytics").
 - Navigation menu with links to different modules (Data Collection, Analysis, Reporting).
- **Main Metrics Section:**
 - Key performance indicators (KPIs) such as:
 - **Attrition Rate:** Percentage of employees leaving the organization.
 - **Average Job Satisfaction:** Average score based on employee feedback.
 - **Total Employees:** Count of active employees in the organization.
- **Navigation Buttons:**
 - Quick access to the Data Collection, Analysis Results, and Reporting modules.
- **Footer:**
 - Links to help resources, FAQs, and contact information for support.

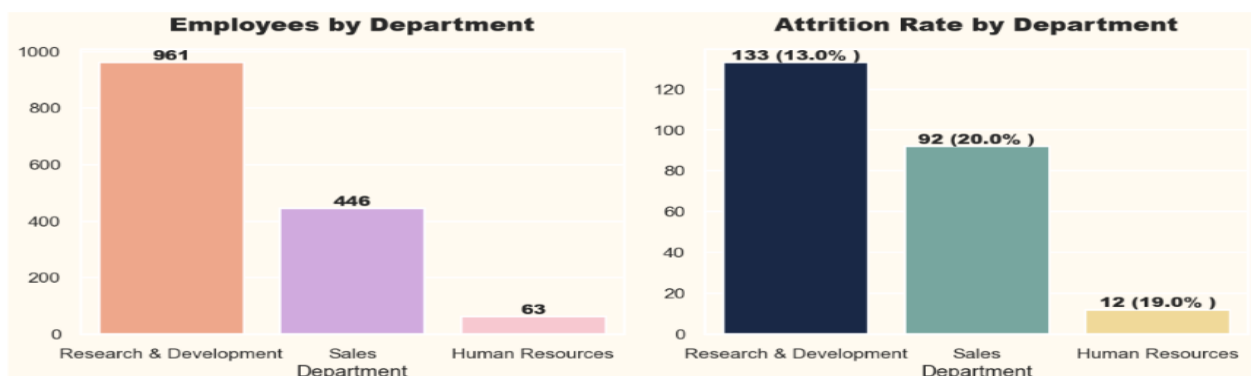


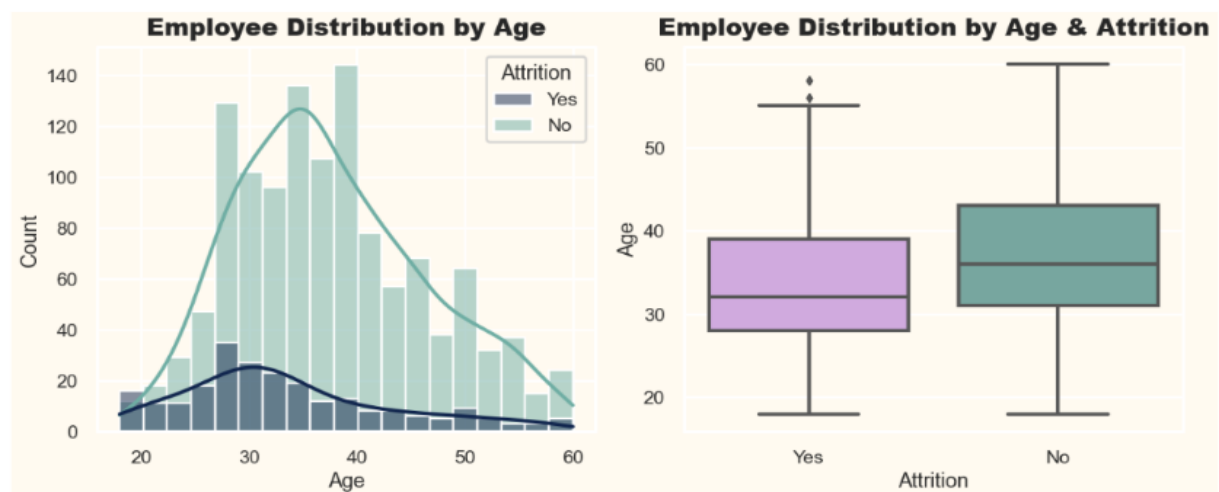
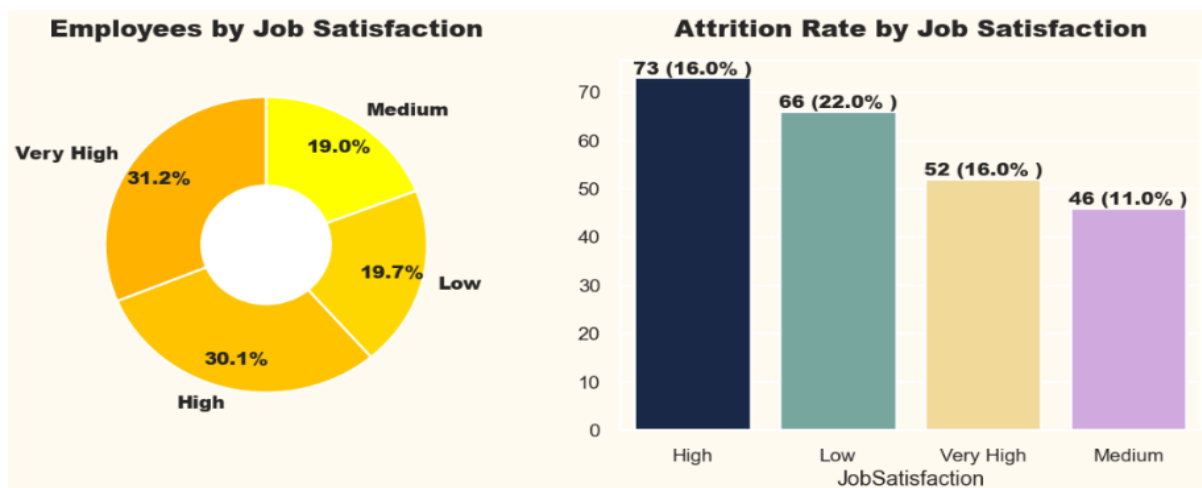
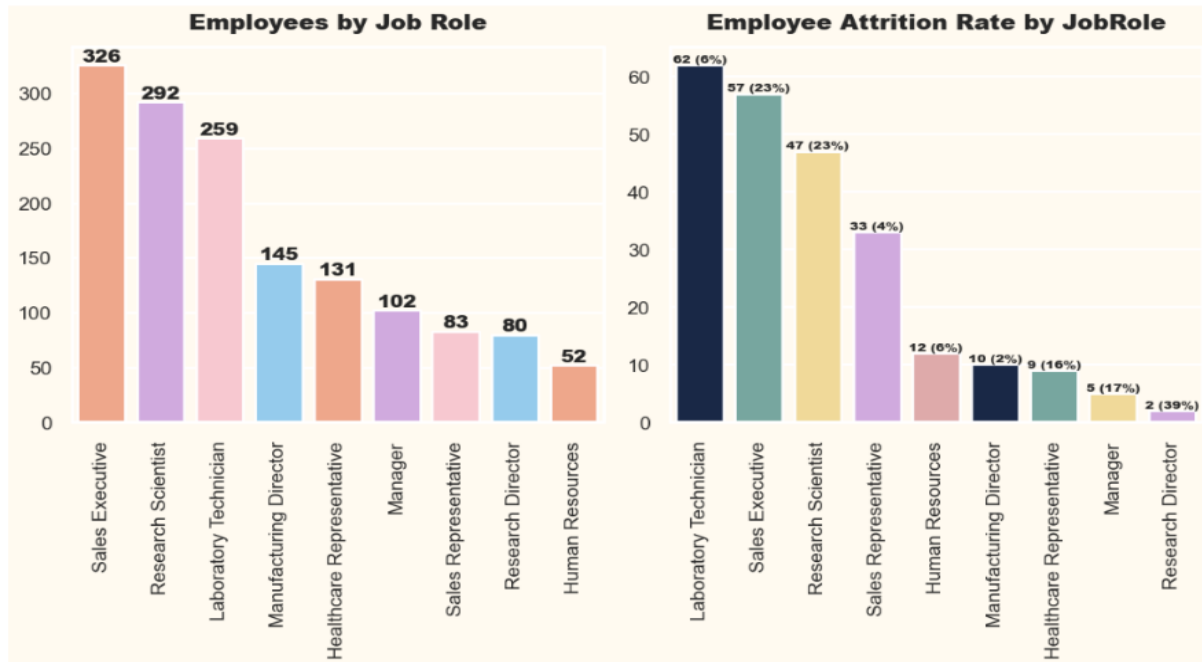
2.2 Data Collection Interface Wireframe

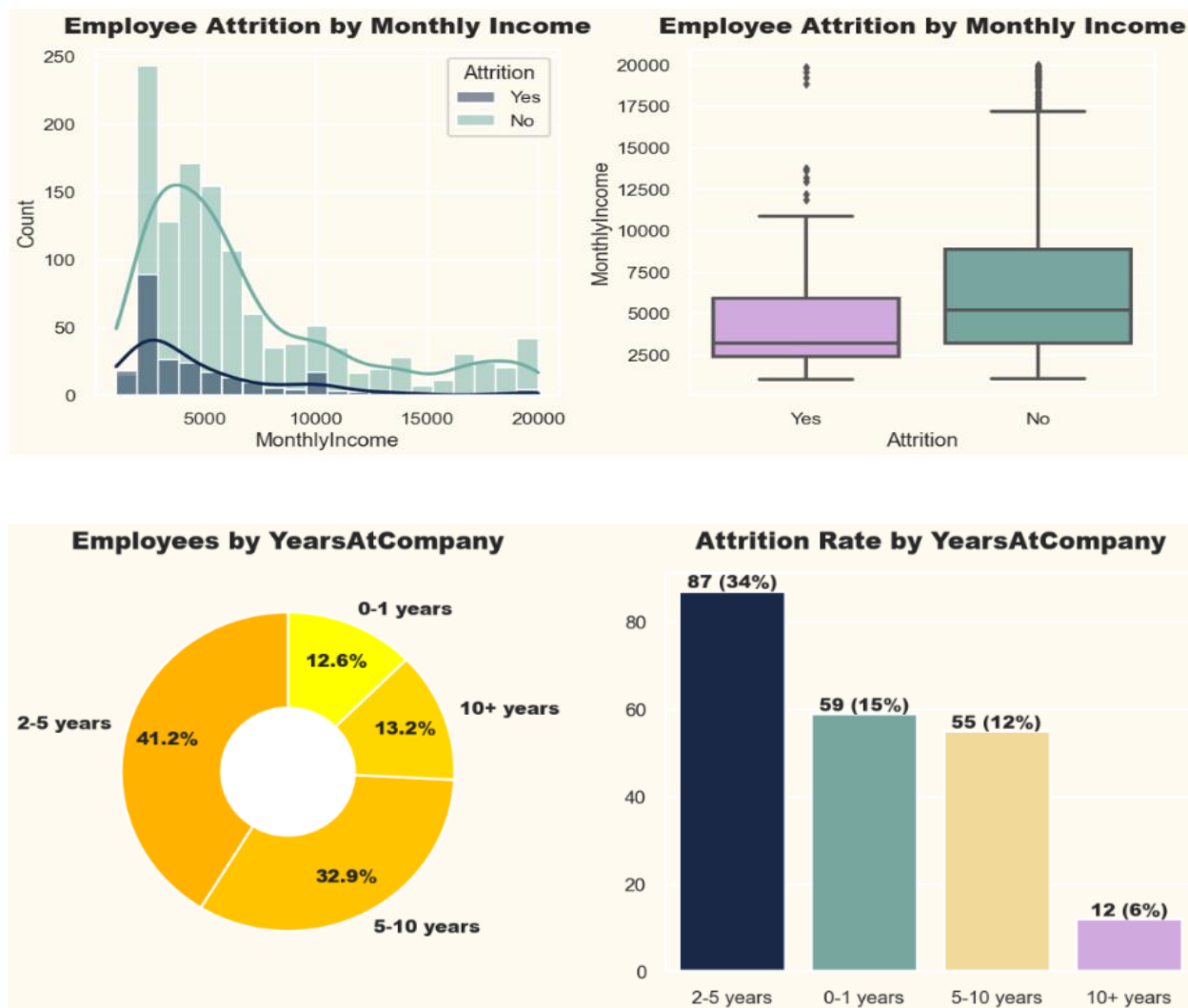
Description: This interface allows users to input employee data for analysis, ensuring data quality and completeness.

Key Elements:

- **Form Fields:**
 - Input fields for employee attributes, including:
 - Name
 - Age
 - Job Role
 - Department
 - Monthly Income
 - Job Satisfaction Score (scale of 1-10)
 - Tenure (in years)
- **Upload Button:**
 - Option to upload bulk data from CSV files, with a sample format for user reference.
- **Submit Button:**
 - To save the entered data, triggering validation checks.
- **Validation Messages:**
 - Notifications for incomplete or incorrect entries (e.g., "Please fill in all required fields" or "Job Satisfaction must be between 1 and 10").







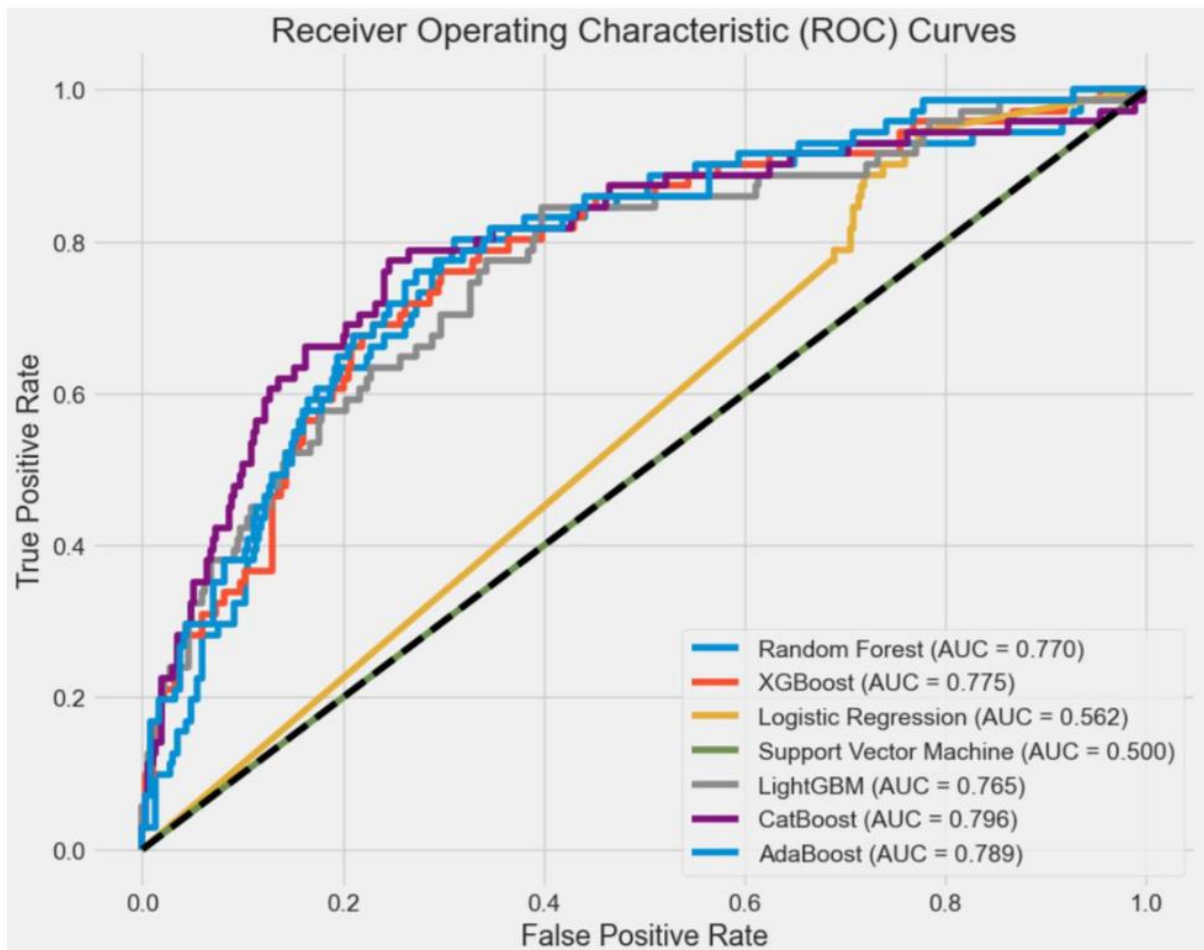
2.3 Analysis Results Page Wireframe

Description: This page displays the results of the analyses conducted on the employee data, providing insights into employee behavior and attrition risk.

Key Elements:

- **Summary Statistics:**
 - Key metrics summarizing the analysis:
 - Total Number of Employees
 - Total Number of Employees at Risk of Attrition
 - Average Job Satisfaction Score

- **Visualizations:**
 - Graphs and charts to illustrate trends and patterns:
 - Bar charts showing attrition rates by department.
 - Line graphs tracking job satisfaction over time.
- **Filter Options:**
 - Dropdown menus to filter results based on various criteria (e.g., department, job role).
- **Export Button:**
 - Option to download the analysis results in PDF or CSV format for reporting purposes.



2.4 Reporting Page Wireframe

Description: The reporting page allows users to generate comprehensive reports based on the analysis results.

Key Elements:

- **Report Generation Options:**
 - Options to select parameters for the report (e.g., time frame, specific metrics).
- **Generated Report Preview:**
 - A section to preview the report before downloading.
- **Download Button:**
 - Option to download the report in various formats (e.g., PDF, Excel).
- **Share Button:**
 - Ability to send reports via email directly from the application.

3. Annotations

3.1 Wireframe Annotations

Each wireframe will include annotations to clarify the functionality of specific elements:

- **Submit Button:**
 - **Functionality:** Validates form fields and saves employee data to the database if validation is successful.
- **Visualization Tools:**
 - **Functionality:** Charts update based on selected filters, providing real-time insights.
- **Export Button:**
 - **Functionality:** Triggers the generation of a downloadable report based on analysis results.

4. User Flow

4.1 User Flow Diagram

A user flow diagram illustrates the path users take through the application. The following steps represent a typical user journey:

1. **Access Home Page:** Users land on the home page to view KPIs and navigate to other modules.
2. **Navigate to Data Collection:** Users click on the Data Collection button to input employee data.
3. **Input Data:** Users enter data or upload a CSV file.
4. **View Analysis Results:** After data submission, users navigate to the Analysis Results page to view insights.
5. **Generate Reports:** Users can generate and download reports based on the analysis.

4.2 Detailed User Flow Steps

- **Step 1:** User opens the application and sees the home page with key metrics.
- **Step 2:** User clicks on the "Data Collection" button to enter employee data.
- **Step 3:** User fills out the form or uploads a CSV, then clicks "Submit."
- **Step 4:** Upon successful submission, the user is directed to the Analysis Results page, displaying insights based on the newly entered data.
- **Step 5:** User selects filters to refine results and reviews visualizations.
- **Step 6:** User clicks on the "Reporting" button to generate a report.
- **Step 7:** User selects parameters and downloads the report in their preferred format.

5. Technical Considerations

5.1 Tools Used

- **Programming Language:** Python
- **Libraries:**
 - **Data Processing:** Pandas, NumPy
 - **Visualization:** Matplotlib, Seaborn
 - **Machine Learning:** Scikit-learn, XGBoost, LightGBM

5.2 Environment Setup

- **Development Environment:** Jupyter Notebook for coding and prototyping.
- **Version Control:** GitHub for maintaining the project's codebase.

6. Security Considerations

6.1 Data Privacy

- **Data Handling:** Ensure compliance with data protection regulations (e.g., GDPR) when handling employee data.
- **Anonymization:** Consider anonymizing sensitive employee information to protect privacy.

6.2 User Authentication

- **Access Control:** Implement user authentication mechanisms to ensure that only authorized personnel can access sensitive data and functionalities.

7. Conclusion

The Wireframe Design Document provides a structured overview of the HR Analytics project's interface. By detailing the layout and functionality of each page, this document serves as a blueprint for the development phase. The comprehensive user flow ensures a seamless experience, helping HR professionals leverage insights into employee attrition and make informed decisions.